

Resolving Forces and Inclined Planes

On a slope, resolve along the plane and perpendicular to the plane. Weight splits into $mg \sin \alpha$ down the slope and $mg \cos \alpha$ into it.



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THE TOPIC ON ONE MAP

Resolving Forces and Inclined Planes

01 — Equilibrium on a slope

parallel and perpendicular

horizontal force

force at an angle

02 — Up or down?

about to slide up

about to slide down

find μ

03 — Moving on a slope

$F = ma$ along the plane

projected up

will it slide back?

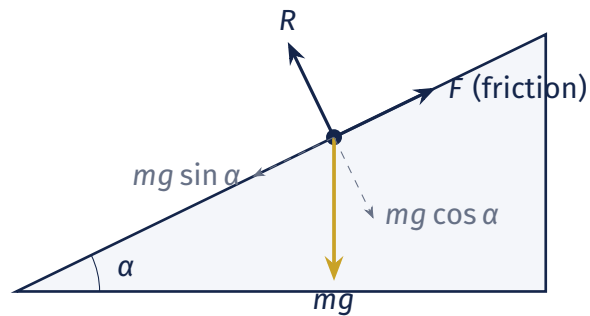
THE SLOPE TRIANGLE

weight: $mg \sin \alpha$ down the slope
slope

$mg \cos \alpha$ into the

sin goes *down* the slope (the steeper, the bigger)
pushes into the plane

$R = mg \cos \alpha$ only if no other force



01 Equilibrium on a Slope: Parallel and Perpendicular

Tilt your axes: one along the slope, one perpendicular. Every force splits into these two. Perpendicular gives R ; parallel gives the balance (and the friction, if any).

WHEN YOU SEE → YOU DO

“inclined at α where $\tan \alpha = \frac{3}{4}$ ”

→ $\sin \alpha = \frac{3}{5}, \cos \alpha = \frac{4}{5}$

a horizontal force X on the slope

→ along: $X \cos \alpha$; into the plane: $X \sin \alpha$

a force at 40° to the plane

→ along: $X \cos 40^\circ$; away from the plane: $X \sin 40^\circ$

“show there is no frictional force”

→ balance along the slope without F

“smallest value of X ”

→ on the point of sliding down: friction up the slope

WORKED MODEL • OCTOBER 2022

2 kg on a plane with $\tan \alpha = \frac{3}{4}$, held by a horizontal force $X = 14.7$ N. Show there is no friction.

1 up the slope: $14.7 \times \frac{4}{5} = 11.76$

$X \cos \alpha$

2 down the slope: $2g \times \frac{3}{5} = 11.76$

$mg \sin \alpha$

ANSWER balanced, so $F = 0$

IDEA 01

Practise: Equilibrium on a Slope: Parallel and Perpendicular

My questions first, then real past papers.

Q1 DR ESLAM Smooth slope, force up the plane

M1 · TOPIC 8 3 marks

A 4 kg particle rests on a smooth plane inclined at 30° , held by a force P up the line of greatest slope. Find P and the normal reaction.

Resolve along the plane and perpendicular to it.

Q2 DR ESLAM Horizontal force on a slope

M1 · TOPIC 8 3 marks

A 2 kg particle rests on a smooth plane inclined at 25° , held by a horizontal force H . Find H .

A horizontal force has a part along the plane ($H \cos 25^\circ$) and a part into the plane.

PAST PAPERS

Q3 PAST PAPER Pearson IAL WME01 · October 2019 · Q4

M1 · TOPIC 8 12 marks

4.

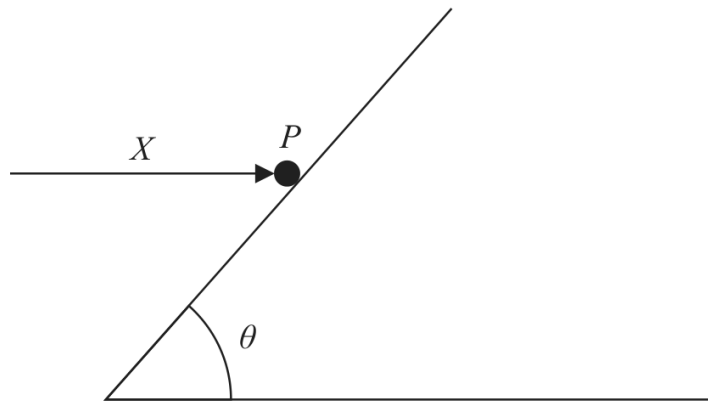


Figure 1

A particle, P , of mass km lies on a fixed rough plane. The plane is inclined to the horizontal at an acute angle θ . A horizontal force of magnitude X acts on P , as shown in Figure 1. The line of action of the force lies in the vertical plane which contains the line of greatest slope of the inclined plane that passes through P . The coefficient of friction between P and the inclined plane is μ .

When $X = mg$, the particle P is in equilibrium and on the point of sliding down the plane.

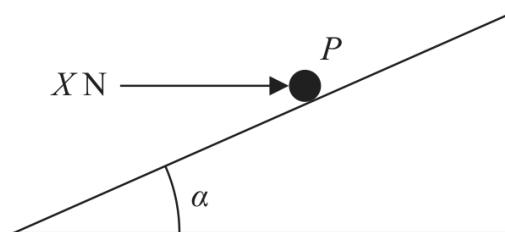
(a) Show that $\mu = \frac{k \tan \theta - 1}{k + \tan \theta}$ (10)

(b) Deduce that, when $k = 1$, θ must be greater than 45° (2)

Q4 PAST PAPER Pearson IAL WME01 · October 2022 · Q3

M1 · TOPIC 8 11 marks

3.

**Figure 2**

A rough plane is inclined to the horizontal at an angle α , where $\tan \alpha = \frac{3}{4}$

A particle P of mass 2 kg is held in equilibrium on the plane by a horizontal force of magnitude X newtons, as shown in Figure 2. The force acts in a vertical plane which contains a line of greatest slope of the inclined plane.

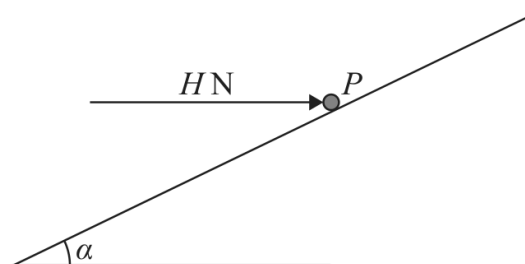
(a) Show that when $X = 14.7$ there is no frictional force acting on P (3)

The coefficient of friction between P and the plane is 0.5

(b) Find the smallest possible value of X . (8)

Q5 PAST PAPER Pearson IAL WME01 · October 2024 · Q6**M1 · TOPIC 8** 12 marks

6.

**Figure 4**

A particle P of mass 5 kg lies on the surface of a rough plane.

The plane is inclined at an angle α to the horizontal, where $\tan \alpha = \frac{3}{4}$

The particle is held in equilibrium by a horizontal force of magnitude H newtons, as shown in Figure 4.

The horizontal force acts in a vertical plane containing a line of greatest slope of the inclined plane.

The coefficient of friction between the particle and the plane is $\frac{1}{4}$

(a) Find the smallest possible value of H .

(6)

The horizontal force is now removed, and P starts to slide down the slope.

In the first T seconds after P is released from rest, P slides 1.5 m down the slope.

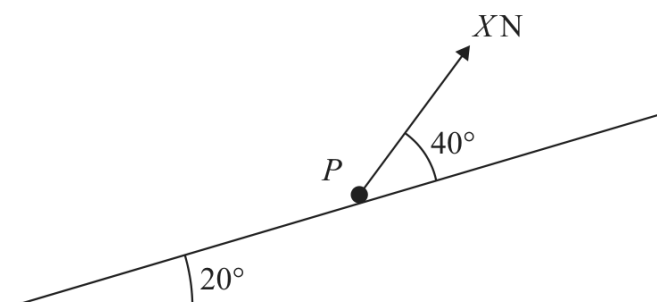
(b) Find the value of T .

(6)

Q6 PAST PAPER Pearson IAL WME01 · June 2026 · Q5

M1 · TOPIC 8 6 marks

5

**Figure 3**

A particle P , whose weight is 50 N , is at rest on a rough plane. The plane is inclined to the horizontal at 20° and the coefficient of friction between P and the plane is 0.3

The particle P is held in equilibrium by a force of magnitude X newtons.

The line of action of the force makes an angle of 40° with the plane, as shown in Figure 3. The line of action of the force lies in a vertical plane which contains a line of greatest slope of the plane.

Given that P is on the point of sliding **down** the plane, find the value of X .

(6)**(Total for Question 5 is 6 marks)**

02 Up or Down? The Two Limiting Cases

Friction always acts against the way the particle is about to move. About to slide **up**: friction points down the slope. About to slide **down**: friction points up the slope. Two situations give two equations.

WHEN YOU SEE → YOU DO

“on the point of sliding up the plane”

→ $F = \mu R$ down the slope

“on the point of sliding down the plane”

→ $F = \mu R$ up the slope

“when $X = 2P$... up; when $X = P$... down”

→ two equations, eliminate P

“the string breaks — does it stay at rest?”

→ compare $\tan \alpha$ with μ

nothing but weight, R , friction

stays at rest when

compare

$\tan \alpha \leq \mu$

WORKED MODEL • JANUARY 2022

$\tan \alpha = \frac{3}{4}$. $X = 2P$: about to slide up; $X = P$: about to slide down. Find μ .

1 up: $2P = mg \sin \alpha + \mu mg \cos \alpha$

friction down

2 down: $P = mg \sin \alpha - \mu mg \cos \alpha$

friction up

3 $0.6 + 0.8\mu = 2(0.6 - 0.8\mu)$

divide the first by the second

ANSWER $\mu = \frac{1}{4}$

IDEA 02

Practise: Up or Down? The Two Limiting Cases

My questions first, then real past papers.

Q7**DR ESLAM**

Does it slide?

M1 · TOPIC 8**3 marks**

A 3 kg block rests on a rough plane inclined at 20° , $\mu = 0.5$. Does it slide? Find the friction.

Q8**DR ESLAM**

Range of a force

M1 · TOPIC 8**4 marks**

A 2 kg particle on a rough plane with $\tan \alpha = \frac{3}{4}$, $\mu = 0.25$, is held by a force P up the plane. Find the range of P for equilibrium.

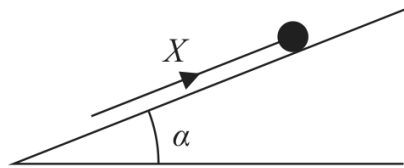
About to slide down: friction up. About to slide up: friction down.

PAST PAPERS

Q9 PAST PAPER Pearson IAL WME01 • January 2022 • Q5

M1 • TOPIC 8 8 marks

5.

**Figure 3**

A particle of mass m rests in equilibrium on a fixed rough plane under the action of a force of magnitude X . The force acts up a line of greatest slope of the plane, as shown in Figure 3.

The plane is inclined at an angle α to the horizontal, where $\tan \alpha = \frac{3}{4}$

The coefficient of friction between the particle and the plane is μ .

- When $X = 2P$, the particle is on the point of sliding up the plane.
- When $X = P$, the particle is on the point of sliding down the plane.

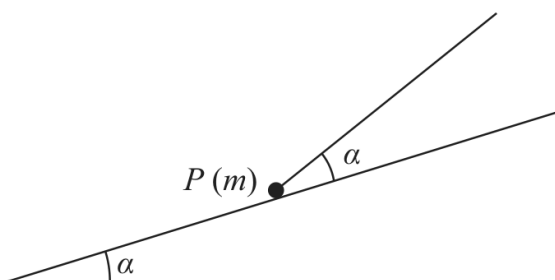
Find the value of μ .

(8)

Q10 PAST PAPER Pearson IAL WME01 · January 2025 · Q6

M1 · TOPIC 8 12 marks

6.

**Figure 3**

A particle P of mass m is held in equilibrium on a fixed rough inclined plane by a light inextensible string.

The plane is inclined at an angle α to the horizontal, where $\alpha < 45^\circ$

The string is inclined to the plane at angle α , as shown in Figure 3.

The string lies in a vertical plane that contains a line of greatest slope of the inclined plane.

When the tension in the string is $0.75mg$, P is on the point of moving up the plane.

- (a) Find an expression for the magnitude of the frictional force acting on P , giving your answer in terms of m , g and α .

(3)

The coefficient of friction between P and the plane is $\frac{1}{2}$

- (b) Show that

$$\tan \alpha = \frac{2}{5}$$

(6)

The string breaks.

- (c) Determine whether P remains at rest. You must justify your reasoning.

(3)

03 Moving on a Slope: $F = ma$ Along the Plane

Moving means friction is μR , against the motion. Resolve perpendicular for R , then write resultant force = ma along the slope. Then SUVAT takes over.

WHEN YOU SEE → YOU DO

“projected up the plane”

→ weight part and friction both down: big deceleration

“slides down the plane”

→ $mg \sin \alpha - \mu R = ma$

“comes to instantaneous rest ... then slides down”

→ new friction direction, new acceleration

“determine whether it remains at rest at C”

→ compare $mg \sin \alpha$ with μR

a pulley at the top of the slope

→ one equation for each particle, same a

R from perpendicular

Friction = μR against motion

$F = ma$ along the slope

SUVAT

WORKED MODEL • JUNE 2025

Projected up a plane with $\tan \theta = \frac{3}{4}$, $\mu = \frac{1}{4}$. Show the deceleration is $\frac{4}{5}g$.

1 $R = mg \cos \theta = \frac{4}{5}mg$; $F = \frac{1}{4} \cdot \frac{4}{5}mg = \frac{1}{5}mg$

perpendicular, then μR

2 $-\frac{3}{5}mg - \frac{1}{5}mg = ma$

both down the slope

ANSWER $a = -\frac{4}{5}g$

× **TRAP** Keeping friction up the slope after the particle turns round. Friction always opposes the motion.

IDEA 03

Practise: Moving on a Slope: $F = ma$ Along the Plane

My questions first, then real past papers.

Q11 DR ESLAM Sliding down

M1 · TOPIC 8 3 marks

A 5 kg particle slides down a rough plane inclined at 30° , $\mu = 0.2$. Find its acceleration.

Q12 DR ESLAM Projected up the slope

M1 · TOPIC 8 4 marks

A particle is projected up a rough plane ($\tan \alpha = \frac{3}{4}$, $\mu = 0.5$) at 8 m s^{-1} . How far up does it go? Will it slide back?

Going up, weight AND friction both act down the plane.

PAST PAPERS

Q13 PAST PAPER Pearson IAL WME01 · June 2021 · Q6

M1 · TOPIC 8 13 marks

6. A fixed rough plane is inclined at an angle θ to the horizontal, where $\tan \theta = \frac{5}{12}$

A particle of mass 6 kg is projected with speed 5 m s^{-1} from a point A on the plane, up a line of greatest slope of the plane.

The coefficient of friction between the particle and the plane is $\frac{1}{4}$

- (a) Find the magnitude of the frictional force acting on the particle as it moves up the plane. (3)

The particle comes to instantaneous rest at the point B .

- (b) Find the distance AB . (5)

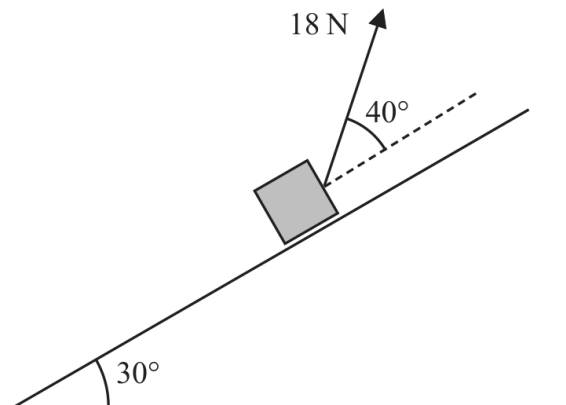
The particle now slides down the plane from B . At the instant when the particle passes through the point C on the plane, the speed of the particle is again 5 m s^{-1}

- (c) Find the distance BC . (5)

Q14 PAST PAPER Pearson IAL WME01 · January 2023 · Q8

M1 · TOPIC 8 15 marks

8.

**Figure 5**

A parcel of mass 2 kg is pulled up a rough inclined plane by the action of a constant force.

The force has magnitude 18 N and acts at an angle of 40° to the plane.

The line of action of the force lies in a vertical plane containing a line of greatest slope of the inclined plane.

The plane is inclined at an angle of 30° to the horizontal, as shown in Figure 5.

The coefficient of friction between the plane and the parcel is 0.3

The parcel is modelled as a particle P

(a) Find the acceleration of P (8)

The points A and B lie on a line of greatest slope of the plane, where $AB = 5$ m and B is above A . Particle P passes through A with speed 2 m s^{-1} in the direction AB .

(b) Find the speed of P as it passes through B . (3)

The force of 18 N is removed at the instant P passes through B . As a result, P comes to rest at the point C .

(c) Determine whether P will remain at rest at C . You must show all stages of your working clearly. (4)

Q15 PAST PAPER Pearson IAL WME01 · January 2024 · Q8

M1 · TOPIC 8 12 marks

8.

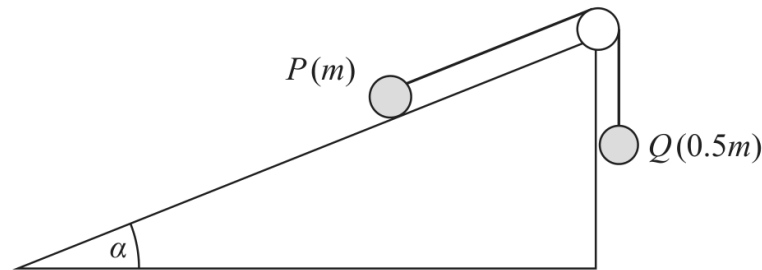


Figure 5

A fixed rough plane is inclined at an angle α to the horizontal, where $\tan \alpha = \frac{5}{12}$

A small smooth pulley is fixed at the top of the plane.

One end of a light inextensible string is attached to a particle P which is at rest on the plane. The string passes over the pulley and the other end of the string is attached to a particle Q which hangs vertically below the pulley, as shown in Figure 5.

Particle P has mass m and particle Q has mass $0.5m$

The string from P to the pulley lies along a line of greatest slope of the plane.

The coefficient of friction between P and the plane is μ .

The system is in **limiting equilibrium** with the string taut and P is on the point of slipping **up** the plane.

(a) Find the value of μ .

(8)

The string breaks and P begins to move down the plane.

When particle P has travelled a distance of 0.8 m down the plane, the speed of P is $V \text{ m s}^{-1}$

(b) Find the value of V .

(4)

Q16 PAST PAPER Pearson IAL WME01 · June 2025 · Q6**M1 · TOPIC 8** 11 marks

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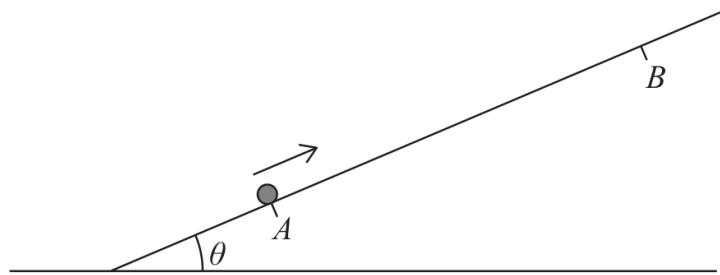
**Figure 4**

Figure 4 shows a rough plane inclined at an angle θ to the horizontal, where $\tan \theta = \frac{3}{4}$

The points A and B lie on a line of greatest slope of the plane, with B above A .

A package P of mass m is projected up the plane from A towards B .

The coefficient of friction between the plane and the package is $\frac{1}{4}$

The package P is modelled as a particle.

(a) Show that the **deceleration** of P , as it moves from A to B , is $\frac{4}{5}g$ (6)

The package P comes to rest at B .

Given that P is projected from A with speed $U \text{ ms}^{-1}$ and that $AB = 1.5 \text{ m}$,

(b) find the value of U . (2)

On reaching B , P is held at rest there by a force of magnitude X newtons acting up the plane in the direction AB .

Given that the mass of P is 2 kg ,

(c) find the smallest possible value of X . (3)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|--------------------|--|
| 01 | Equilibrium | Axes along and perpendicular to the plane. |
| 02 | Up or down | Friction opposes the slide; two cases, two equations. |
| 03 | Moving | R , then μR against the motion, then $F = ma$. |

BOTTOM LINE $mg \sin \alpha$ down the slope · $mg \cos \alpha$ into it · friction against the motion · stays at rest if $\tan \alpha \leq \mu$.

Answers

Resolving Forces and Inclined Planes · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · EQUILIBRIUM ON A SLOPE: PARALLEL AND PERPENDICULAR

- Q1** Dr Eslam · Smooth slope, force up the plane $P = 4g \sin 30^\circ = 19.6 \text{ N}; R = 4g \cos 30^\circ \approx 33.9 \text{ N}$
- Q2** Dr Eslam · Horizontal force on a slope $H \cos 25^\circ = 2g \sin 25^\circ \Rightarrow H \approx 9.14 \text{ N}$
- Q3** Pearson IAL WME01 · October 2019 · Q4 (a) $\mu = \frac{k \tan \theta - 1}{k + \tan \theta}$ (b) $\mu > 0 \Rightarrow \tan \theta > 1 \Rightarrow \theta > 45^\circ$
- Q4** Pearson IAL WME01 · October 2022 · Q3 (a) $X \cos \alpha = 2g \sin \alpha \Rightarrow X = 14.7$: no friction needed (b) $X = \frac{4g}{11} \approx 3.56 \text{ N}$
- Q5** Pearson IAL WME01 · October 2024 · Q6 (a) $H = \frac{40g}{19} \approx 20.6 \text{ N}$ (b) $T = \sqrt{\frac{15}{2g}} \approx 0.875 \text{ s}$
- Q6** Pearson IAL WME01 · June 2026 · Q5 $X \approx 5.24$

IDEA 02 · UP OR DOWN? THE TWO LIMITING CASES

- Q7** Dr Eslam · Does it slide? down-slope part $10.1 \text{ N} < \mu R = 13.8 \text{ N}$: it stays; $F = 10.1 \text{ N}$ up the plane
- Q8** Dr Eslam · Range of a force $7.84 \leq P \leq 15.68 \text{ N}$
- Q9** Pearson IAL WME01 · January 2022 · Q5 $\mu = \frac{1}{4}$
- Q10** Pearson IAL WME01 · January 2025 · Q6 (a) $F = 0.75mg \cos \alpha - mg \sin \alpha$ (b) $\tan \alpha = \frac{2}{5}$
(c) $\tan \alpha < \mu$: P remains at rest

IDEA 03 · MOVING ON A SLOPE: $F = ma$ ALONG THE PLANE

- Q11** Dr Eslam • Sliding down
 $a = g(\sin 30^\circ - 0.2 \cos 30^\circ) \approx 3.20 \text{ m s}^{-2}$
- Q12** Dr Eslam • Projected up the slope
 deceleration $g(0.6 + 0.4) = 9.8$: $s \approx 3.27 \text{ m}$;
 $\tan \alpha > \mu$, so it slides back
- Q13** Pearson IAL WME01 • June 2021 • Q6
 (a) $F \approx 13.6 \text{ N}$ (b) $AB \approx 2.07 \text{ m}$ (c) $BC \approx 8.29 \text{ m}$
- Q14** Pearson IAL WME01 • January 2023 • Q8
 (a) 1.18 m s^{-2} up the plane (b) 3.98 m s^{-1} (c)
 $\tan 30^\circ > 0.3$: it does not stay at rest
- Q15** Pearson IAL WME01 • January 2024 • Q8
 (a) $\mu = \frac{1}{8}$ (b) $V \approx 2.05$
- Q16** Pearson IAL WME01 • June 2025 • Q6
 (a) $g(\frac{3}{5} + \frac{1}{4} \cdot \frac{4}{5}) = \frac{4}{5}g$ (b) $U = \sqrt{\frac{12g}{5}} \approx 4.85$ (c)
 $X = \frac{4g}{5} \approx 7.84 \text{ N}$